Drawing DAGs: trees and heat-related illness

Is urban forestry public health?

In his piece, How Public Health Took Part in Its Own Downfall, Ed Yong makes the case that public health should be viewed quite broadly. He didn't say this, but I think you could extend his argument to the idea that urban forestry, the planting of trees in towns and cities, could be viewed as an aspect of public health. Excess heat can cause mortality directly and indirectly, as well as simply exacerbate existing health issues. The several other readings about trees discuss their importance in reducing heat for local communities as well as their unequal distribution. Tim Hartford's story about a Chicago heatwave emphasized the importance of trust and community in responding to heat waves. And of course Given all of this, I would like you to…

Draw a DAG about the factors that influence heat-related illness and deaths

Draw graphs showing the relationships

Each arrow represents some causal relationship between the variable upstream and that downstream. The arrows, however, do not represent any particular shape to that relationship (e.g., increasing, decreasing, sigmoidal, accelerating, hump-shaped).

I would like you to draw a graph showing the hypothetical relationship between the upstream variable (x-axis) and the downstream variable (y-axis). If there are other factors that cause that downstream variable and *modify* the relationship between the upstream and downstream variable (e.g., another arrow into the downstream arrow[1] that interacts with the upstream-downstream relationship), go ahead and account for that with different lines (e.g., one for low income, one for high income). The more specific you can make these graphs, the better.

[1] This is getting tedious… I should have used x and y!

Use your DAG as a tool

Now, imagine you are responsible for the public health of a metropolitan area that is expected to face more and more severe heat waves. Given your understanding of causation in this system, which 1–2 things would you try to change? Why?

Based on your DAG and graphs, what advantages or problems might you face?